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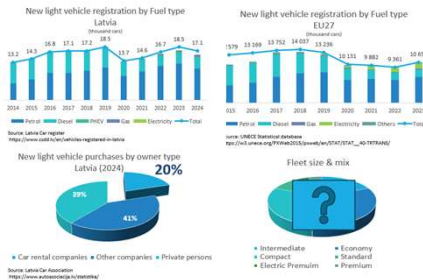
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# Introduction

### Problem definition:

The growing focus on urban mobility and the development of fleet telematics have accelerated the growth of car rental services;

Imbalances between fleet demand and supply present a critical challenge for the capital-intensive car rental industry, where fleet acquisition must be planned in advance;



Even small inaccuracies in rental day projections can lead to significant financial losses;

There is a lack of monthly demand forecasting models validated on real-world data.

### Study Goal:

- develop monthly demand forecasting pipeline for station-based multi service car rental fleet planning, with the objective of accurate planning for fleet size and mix.

### Research Questions:

- What are the machine learning models and unsolved gaps in monthly fleet demand forecasting in mid-term planning?
- How fleet segmentation affect the predictive performance of monthly demand forecasting using different machine learning models?
- Which machine learning algorithms generate most accurate forecasting monthly fleet demand?
- How do proposed models perform on different time frames?

### Objectives:

- To analyze the evolution and current landscape of car fleet forecasting by identifying existing solutions and research gaps up to date.
- To develop a forecasting pipeline using advanced machine learning techniques with multivariate exogenous inputs.
- To ensure model performance through data preprocessing and fleet segmentation.
- To provide accurate mid-term demand forecast for fleet size and vehicle mix planning taking into account limitation of computational resources.
- To validate forecasting pipeline using real world car rental data and industry expert feedback.

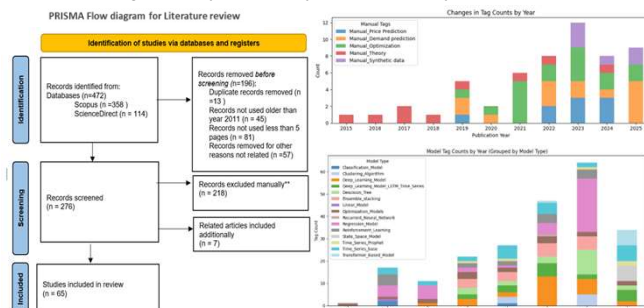
## Methodology

### Study Design:

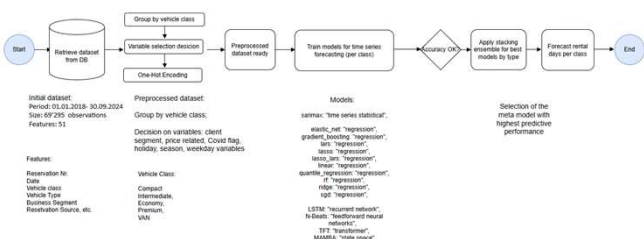
- A quantitative research design combining systematic literature review and empirical data analysis to examine machine-learning approaches for car rental days forecasting. The study utilizes Python programming language.

### Materials and Procedures:

- Literature review was conducted following PRISMA guidelines, with studies categorized by problem type and model type.



### Forecasting pipeline:



## Results and Discussion

### Results:

- **225 forecasting models evaluated:** 15 models  $\times$  3 horizons (week, month, quarter)  $\times$  5 vehicle classes, with additional experiments conducted for hyperparameter tuning and stacking ensemble for the best-performing meta-model.
- **Graphs and Images:** Model comparison plots for 7, 30, and 92 days, Vehicle-class, specific visualizations, summary tables.
- **Evaluation metric:** RMSE, measured in rental days as target variable.

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### Key Findings:

- Forecast accuracy is driven by demand density and volatility, rather than forecast horizon length.
- Stable classes (Economy, Premium) show similar performance across models, except LARS. Economy: 30d- RF 10.0, GB 9.5; 92d- RF 18.6, GB 19.7, LSTM 25.4. Premium: 30d & 92d- GB, RF, LSTM 1.0-2.0 level.
- Volatile classes (Compact, Intermediate) show large performance gaps.
- Compact (92d): GB 68.8, LSTM 77.1, TFT 98.3. Intermediate (92d): GB 34.2, LSTM 64.4, Mamba 67.9, TFT 52.4.
- Nonlinear regression models Gradient Boosting (GB), Random Forest (RF) capture volatile demand structures more effectively.
- Deep learning models are more sensitive to noise and structural complexity; sequence-based models struggle with irregular demand and strong seasonality at long horizons.
- LSTM provides the best accuracy–cost trade-off, N-BEATS is an efficient short- to medium-term alternative, while Temporal Fusion Transformer (TFT) performs well on monthly horizon, but do not capture seasonality when forecast horizon increase and was computationally expensive.
- The GB stacking ensemble overfits under low-sample conditions.

**Implication:**

- Apply class-specific model selection and careful variable analysis.
- Prefer robust, interpretable models under real-world constraints, as performance depends more on demand signal characteristics than model complexity.

**Limitations:**

- Dataset cannot be publicly released due to confidentiality constraints.
- COVID-19 impacts introduce structural breaks patterns in the data.
- Computational constraints limit extensive tuning of deep learning models.

## Conclusions / Outcomes

- ✓ **Validated pipeline:** A monthly forecasting pipeline was developed and validated on real-world car rental data for fleet and vehicle-mix planning.
- ✓ **Demand-driven model choice:** No single model dominates; performance depends on volatility, density, and structure.
- ✓ **Industry relevance:** Executive feedback from Baltic and Finnish markets confirms real-world applicability.

## Key References

- Oliveira, B. B., Carravilla, M. A., & Oliveira, J. F. (2017). Fleet and revenue management in car rental companies: A literature review and an integrated conceptual framework. *Omega*, 71, 11–26. <https://doi.org/10.1016/j.omega.2016.08.011>
- Lazov, I. (2017). Profit management of car rental companies. *European Journal of Operational Research*, 258(1), 307–314. <https://doi.org/10.1016/j.ejor.2016.08.064>

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- Dataset provider: SIA Transparent (subject to data availability and confidentiality constraints).

## Contacts

LinkedIn

